

SAI Happy Farms Farm ERP — Administrator User Manual

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1. Admin Role & Dashboard

The Administrator holds absolute control over the ERP system. While Farmers and Supervisors interact with limited datasets, the Admin has global read/write access.

The Enterprise Analytics Dashboard functions identically for Admins as it does for Supervisors, but with zero farm restrictions—you can view and filter data for every farm in the organization.

2. User Management

Located via the `AdminUsersPage` interface, this module leverages the Node.js Express backend API (using the Firebase Admin SDK) to bypass standard restrictions.

Creation and Assignment

- Roles: You can create users and assign them the roles of `Farmer`, `Supervisor`, or `Admin`.
- Farm Mapping: A user's `farmIds` array dictates their access. You map farmers to a specific farm, and supervisors to an array of farms.

- Status Toggles: An active toggle allows you to instantly revoke application access without deleting the user.

Password Requirements

- Strict Validation: Passwords for ALL roles (Admin, Supervisor, Farmer) MUST be exactly 8 characters or more. Submitting a 7-character password will be rejected by the validation middleware.

(Note: True hard-deletion functionality of user records is not verified from the available project evidence; use the Active status toggle to disable accounts).

3. Farm & Master Inventory Management

Located in the `AdminFarmsPage`, this module controls the single source of truth for the organization.

- Master Records: The `currentBirdCount` and `currentFeedKg` act as the master ledgers.
- Atomic Integrity: When a farmer submits a daily report, Firestore utilizes atomic transactions to deduct mortality and feed from this master record.

4. Prediction Engine

The system features an automated forecasting model located on the `PredictionPage`.

- Purpose: To forecast egg production output over the next 14, 21, or 30 days.
- Logic: The engine utilizes a Weighted Moving Average (WMA) alongside Linear Regression, blended against hardcoded standard breeder curves (`CF\_STD` and `FR\_STD`).
- Limitations: The engine requires sufficient historical data to establish an accurate trend line.
- Important: Predictions are statistical forecasts, not AI-guaranteed outcomes. Do not present them to stakeholders as absolute guarantees.

5. Historical Data Import (XLSX, CSV, XML)

Note: The precise functional execution rate for import tests is not verified from the available project evidence, but the workflow is defined per the requirements.

Admins can initiate batch imports of legacy data to bootstrap the system.

Workflow

4. Upload: Select an XLSX, CSV, or XML file.
5. Parsing & Mapping: The system parses columns and maps them to database fields.
6. Farm Matching: The system attempts to match rows to existing `farmId`s.

- ***Strict Rule:*** Unknown or ambiguous farm mappings are explicitly rejected. The system will NOT silently guess assignments.

4. Conflict Handling: The system identifies duplicate records.

5. Execution: The batch imports and logs an entry in the Import History.

6. Import Revert / Rollback Procedure

If a batch import executes with severe mapping errors or unintended data pollution, Admins can execute a total rollback.

The Exact Verified Workflow:

7. Navigate to the Import History.
8. Select the specific batch that needs rolling back.
9. Open the Revert Entire Import Batch dialog.
4. Read the destructive warning notice.
5. Check the agreement box. (If left unchecked, the Confirm button remains completely disabled).
6. Enter the EXACT Batch ID in the text field. The system supports case-insensitive exact matching and is robust against trailing/leading spaces. (Compatible with both `id` and `batchId` properties).
7. The Confirm button will ONLY become enabled when ***both*** the ID matches perfectly and the agreement is checked.
8. Execute the Revert. The batch is pulled entirely from Firestore.

7. Security & Infrastructure Controls

As the Admin, you rely on several automated backend protections:

- BOLA Prevention: Firestore Security Rules strictly mathematically block unauthorized reads/writes to unassigned farms. (Verified by 179 passing security tests).
- Error Sanitization: The Express backend intercepts unhandled 500 errors and strips out internal stack traces, ensuring infrastructure details are never leaked to the client.
- Log Redaction: Passwords and API tokens are automatically scrubbed from server logs.

8. Admin Testing Checklist

Test ID	Module	Steps	Expected Result	Pass/Fail
ADM-001	User Validation	Attempt to create a user with a 6-character password.	Validation explicitly rejects request.	
ADM-002	Rollback Safety	Open Revert dialog. Leave agreement unchecked. Type correct ID.	Confirm button strictly remains disabled.	
ADM-003	Rollback Execute	Check agreement, type exact Batch ID.	Confirm button enables. Execution removes batch.	
ADM-004	Prediction Engine	Run prediction on a newly created farm (0 days history).	Graceful handling of insufficient data.	
ADM-005	Security Audit	Confirm Admin dashboard can read all farms regardless of explicit assignment list.	Success. Global visibility confirmed.	

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